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Expt.No: 5

Title: Python program to perform Basic operations on an Image¶

(i) Chaging Contrast and Brightness (ii) Edge Detection, Blur, Sharpening

In [4]:

```
import cv2
import matplotlib.pyplot as plt
import numpy as np

image = cv2.imread('ResizeImage3.jpg')

adjusted_image = cv2.convertScaleAbs(image, alpha=1.5, beta=50)

# (ii) Edge Detection
edges = cv2.Canny(cv2.cvtColor(image, cv2.COLOR_BGR2GRAY), 100, 200)

# (iii) Apply Gaussian Blur
blurred_image = cv2.GaussianBlur(image, (21, 21), 0)

# (iv) Sharpen the image using a kernel
sharpened_image = cv2.filter2D(
    image,
    -1,
    np.array([[-1, -1, -1],
              [-1,  9, -1],
              [-1, -1, -1]])
)

plt.figure(figsize=(12, 8))

plt.subplot(2, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')

plt.subplot(2, 3, 2)
plt.imshow(cv2.cvtColor(adjusted_image, cv2.COLOR_BGR2RGB))
plt.title('Adjusted Image (Contrast & Brightness)')
plt.axis('off')

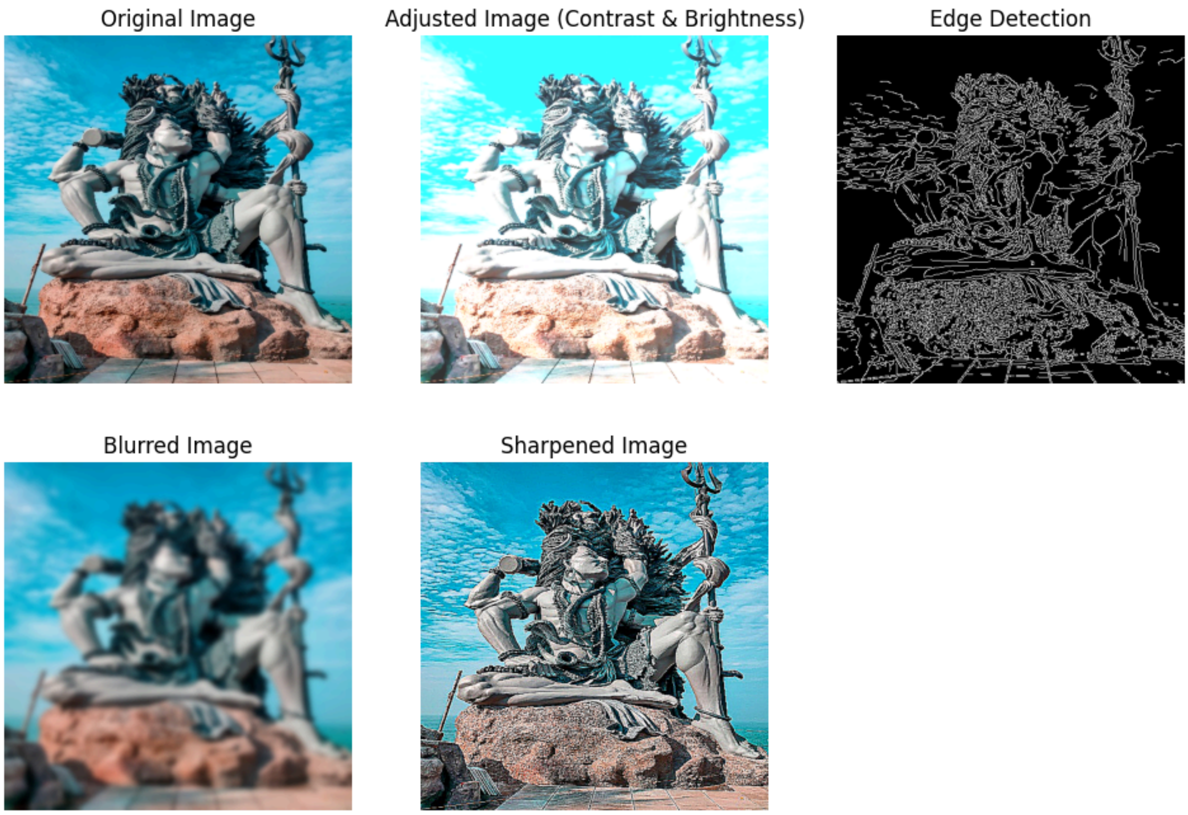
plt.subplot(2, 3, 3)
plt.imshow(edges, cmap='gray')
plt.title('Edge Detection')
plt.axis('off')

plt.subplot(2, 3, 4)
plt.imshow(cv2.cvtColor(blurred_image, cv2.COLOR_BGR2RGB))
plt.title('Blurred Image')
plt.axis('off')

plt.subplot(2, 3, 5)
plt.imshow(cv2.cvtColor(sharpened_image, cv2.COLOR_BGR2RGB))
plt.title('Sharpened Image')
plt.axis('off')

plt.show()

cv2.imwrite('Adjusted_Image.jpg', adjusted_image)
cv2.imwrite('Edge_Detection_Image.jpg', edges)
cv2.imwrite('Blurred_Image.jpg', blurred_image)
cv2.imwrite('Sharpened_Image.jpg', sharpened_image)
```



Out[4]: True

In [ ]:
```
1
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